

Dao Xuan Tan

Software Engineer, AI Systems

📍 Ho Chi Minh, Vietnam

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🌐 Đào Xuân Tân

Summary

Software engineer building high-performance, GPU-backed AI systems on Linux. Production computer vision and deep learning with PyTorch (SAM, Grounding DINO, VLM, LoRA), on a systems core of concurrency, memory management, and profiling; cut peak memory from 200+ GB to under 20 GB and runtime from minutes to under 10 seconds. Strong CS fundamentals, with C++ from coursework and competitive programming, and quick to pick up new languages.

Experience

Pi Associates, Software Engineer, Compute Platform

Ho Chi Minh

Python, asyncio, multiprocessing, PostgreSQL, Kafka, Redis, Docker, Kubernetes, Linux, GCP

Jan 2026 – present

7 months

- Main developer of an internal **Python** compute platform running **tens of thousands** of jobs a day for about **20** researchers and engineers
- Diagnosed and fixed a critical **memory leak**, cutting peak memory from **200+ GB to under 20 GB**
- Cut runtime from **minutes to under 10 seconds** on **1-5 GB** of data per run through **parallelism**, vectorization, and algorithmic improvements
- Designed the **concurrent execution model** (**asyncio** plus **process pool**, GIL-aware) to maximize **throughput** under heavy parallel load
- Built the **PostgreSQL** schema, **Redis** caching, and a **Kafka** producer streaming signals to downstream consumers
- Ran jobs as Dockerized **Kubernetes** workloads across **GCP** and on-premise **Linux**, with **GitHub Actions** CI/CD and **Grafana**

AcamarAI, AI Engineer

Ho Chi Minh

Python, PyTorch, computer vision, SAM, Grounding DINO, VLM, LoRA, vLLM, FastAPI

June 2025 – Jan 2026

8 months

- Applied **computer vision** in production, object segmentation with **SAM** and **Grounding DINO**, **ColPali** vision-RAG, and **VLM** document understanding
- Fine-tuned task-specific models with **LoRA/PyTorch** and prepared datasets to improve accuracy for target tasks (**deep learning**)
- Served self-hosted **Qwen** models with **vLLM** on a single **A100 80GB** behind an OpenAI-compatible API, tuning **GPU inference** to about **1s TTFT** and **200 tok/s** aggregate throughput
- Delivered **3 of 14** core AI features end to end, from requirement to production, for **Fortune 500** beta users
- Integrated models into scalable back-end services with async **FastAPI** and **SSE** streaming, schema-validated through **Instructor/Pydantic**, with **evaluation** harnesses (**recall@k**, **LLM-as-judge**)

Skills

Programming: Python (asyncio, multiprocessing), C++ (coursework, competitive programming), SQL, algorithms and data structures

Computer Vision & Deep Learning: PyTorch, SAM, Grounding DINO, object segmentation, ColPali, VLM, LoRA fine-tuning

Concurrency & Performance: asyncio, multiprocessing, GIL-aware concurrency, memory profiling and leak debugging

GPU & Inference: vLLM self-hosted serving on A100, low-latency and high-throughput serving, CUDA kernel basics (self-study)

Backend & Distributed Systems: FastAPI, async APIs, SSE streaming, PostgreSQL, Redis, Kafka, pandas, data pipelines

Linux & DevOps: Linux, Docker, Kubernetes, GCP and on-premise, GitHub Actions CI/CD, Grafana, pytest

LLM Systems: RAG, Qdrant vector search, tool-calling agents, LangChain/LangGraph, Instructor/Pydantic validation

Languages: Vietnamese (Native), English (IELTS 6.5)

Education

Ho Chi Minh University of Science, Mathematics & Computer Science (Data Science)

Ho Chi Minh City

- GPA: 8.74/10.0 (3.74/4.0)

Oct 2022 – Sept 2026